

# HEALTHCARE COST ESTIMATOR

---

## Final Documentation Report

**Project Report Submitted for Internship Project Submission**

### Submitted By

**Name:** Shivam Subedar Patel

**Internship Role:** Intern (AI/ML)

**Project Title:** Healthcare Cost Estimator

### Technology Stack:

Python | Machine Learning | Regression Modeling | Scikit-learn | XGBoost | Streamlit

### Project Description:

A machine learning regression system developed to estimate annual healthcare insurance costs from demographic, lifestyle, and regional attributes. The project includes data preprocessing, exploratory analysis, model comparison, feature importance, and an interactive Streamlit dashboard with prediction and generated-plot pages.

### Submitted To

**Company/Organization Name:** Data Vidwan

**Mentor/Manager Name:** Meet Mistry

**Date of Submission:**

11/06/2026

# Healthcare Cost Estimator - Final Documentation Report

---

## 1. Project Objective

The objective of this project is to build a machine learning regression model that predicts healthcare insurance costs using age, BMI, smoking status, gender, number of dependents, and region. The project also presents model outputs through a professional Streamlit application with dashboard insights, prediction, plots, and generated data views.

## 2.Requirement Coverage

| Requirement                                | Status    | Project File / Output                                    |
|--------------------------------------------|-----------|----------------------------------------------------------|
| Healthcare Cost Prediction Model           | Completed | models/healthcare_cost_model.pkl                         |
| Cleaned and Processed Dataset              | Completed | data/insurance_cleaned.csv, data/insurance_processed.csv |
| Training and Testing Dataset               | Completed | data/train.csv, data/test.csv                            |
| Exploratory Data Analysis Report           | Completed | reports/EDA_Report.docx                                  |
| Regression Model Evaluation Report         | Completed | reports/Model_Evaluation_Report.docx                     |
| Cost Estimation Interface                  | Completed | app/app.py                                               |
| Feature Importance / Explainability Report | Completed | reports/Feature_Importance_Report.docx                   |
| Healthcare Cost Analysis Dashboard         | Completed | app/app.py                                               |
| Project Documentation                      | Completed | reports/Healthcare_Cost_Estimator_Documentation.docx     |

## 3. Dataset

Dataset: Medical Insurance Cost Dataset

Source: Project dataset file data/insurance.csv

Raw records: 1,338

Cleaned records: 1,337

Processed records: 1,337

Training records: 1,069

Testing records: 268

The dataset contains age, sex, BMI, children/dependents, smoker, region, and charges. The target variable is charges, representing annual insurance cost in USD.

## 4. Work Completed

- Loaded and inspected the raw insurance dataset.
- Removed duplicate records and prepared a cleaned dataset.
- Encoded categorical variables for gender, smoking status, and region.
- Created a BMI-age interaction feature for model training.
- Generated training and testing CSV files from the processed dataset.
- Performed exploratory data analysis for age, BMI, smoking impact, region, cost distribution, and correlations.
- Trained and compared Linear Regression, Random Forest Regressor, XGBoost Regressor, and Tuned XGBoost Regressor.
- Selected the best model using actual evaluation metrics.
- Exported the trained model and feature schema.
- Generated EDA, model evaluation, feature importance, and final documentation reports.
- Developed a Streamlit application with dashboard insights, prediction, and plots/data pages.

## 5. Methodology

### 5.1 Data Cleaning

The raw dataset was cleaned by removing duplicate records and preserving valid input columns required for healthcare cost prediction.

### 5.2 Data Preprocessing

Categorical columns were converted into numerical model features using one-hot encoding. The processed dataset stores the encoded feature columns and the target charges column.

### 5.3 Feature Engineering

A BMI-age interaction feature was added to capture combined age and BMI effects in the regression model.

### 5.4 Model Training and Selection

Multiple regression models were compared using MAE, RMSE, and R2 score. The final model was selected using the highest R2 score and supporting error metrics from the generated comparison table.

## 6. Model Results

| Model             | MAE      | RMSE     | R2     |
|-------------------|----------|----------|--------|
| Linear Regression | 4,176.92 | 5,956.45 | 0.8069 |
| Random Forest     | 2,619.61 | 4,665.05 | 0.8816 |
| XGBoost           | 2,592.25 | 4,456.00 | 0.8919 |
| Tuned XGBoost     | 2,444.16 | 4,229.80 | 0.9026 |

## 7. Selected Model

The selected model is Tuned XGBoost. It was selected because it achieved the highest recorded R2 score among the tested regression models.

Exported model file: models/healthcare\_cost\_model.pkl

Feature schema file: models/feature\_names.pkl

Selected model metrics: MAE=2,444.16, RMSE=4,229.80, R2=0.9026.

## 8. Streamlit Application

The Streamlit application is implemented in app/app.py. It includes the following pages:

- Dashboard Insights
- Healthcare Cost Prediction
- Plots and Data

The dashboard page shows dataset-level insights, cost drivers, regional summaries, best model metrics, and feature importance. The prediction page keeps the existing cost estimation interface. The plots and data page displays the generated EDA visualizations and cleaned dataset preview.

## 9. Key Findings

The cleaned dataset contains 1,337 insurance records.

The average annual charge in the cleaned dataset is \$13,279.12, and the median annual charge is \$9,386.16.

The highest average cost by smoking status is for yes records, with an average cost of \$32,050.23.

The highest average regional cost is in the southeast region, with an average cost of \$14,735.41.

The top model feature by saved importance value is smoker\_yes.

## 10. Cost Analysis Recommendations

- Use smoking status and BMI-related insights as major review points when interpreting estimated costs.

- Use regional summaries to compare broad cost variation across available regions.
- Use predictions as indicative estimates only, not as final billing or insurance approval decisions.
- Continue monitoring model performance if new datasets or additional healthcare attributes become available.

## 11. Achievements

- Completed an end-to-end healthcare cost regression workflow.
- Built and exported a reusable trained model artifact.
- Prepared cleaned, processed, training, and testing datasets.
- Created a professional Streamlit application with dashboard, prediction, and data visualization pages.
- Prepared final documentation and supporting DOCX reports.

## 12. Limitations

- The dataset is limited to the available insurance attributes and does not include medical history, plan type, income, provider network, or claim history.
- The model output is an estimate based on historical dataset patterns and should not be treated as a medical, clinical, or official insurance decision system.
- The target charges are recorded in USD, so any converted display value is only for readability.

## 13. Final Submission Contents

The final submission should contain:

- app/app.py
- requirements.txt
- README.md
- data/
- models/
- notebooks/
- reports/
- Documentation/

## 14. How to Run

Install dependencies:

```
pip install -r requirements.txt
```

Run the Streamlit app:

```
python -m streamlit run app/app.py
```

## **15. Final Conclusion**

The Healthcare Cost Estimator project successfully developed an end-to-end regression workflow for estimating annual healthcare insurance costs. The project involved data cleaning, preprocessing, feature engineering, exploratory data analysis, model training, model evaluation, and feature importance analysis.

Different regression approaches were compared, and Tuned XGBoost was selected based on the recorded evaluation results. The saved model is integrated into the Streamlit application for interactive cost estimation.

The final application provides dashboard insights, a prediction interface, and a generated plots/data page while maintaining a consistent professional design. The supporting reports and documentation organize the project outputs for internship submission.